

two deconvolutional layers. During the training, the authors used asymmetric convolutions [43, 15] to enhance the kernel skeleton, which were fused after training. The model was trained on 256×256 px patches, *Charbonnier* loss was used as a target fidelity metric. The model parameters were optimized using Adam with a batch size of 16 and an initial learning rate of $1e - 4$ decreased to $1e - 7$ using the cosine annealing strategy. It should be mentioned that the original model was trained in PyTorch and then exported to TensorFlow and TFLite formats.

4.6. GdAlgo

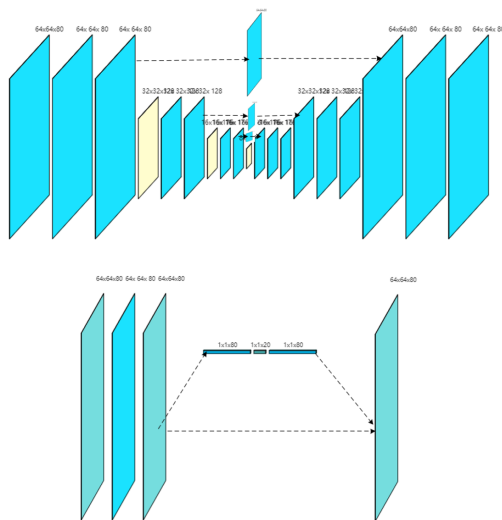


Figure 9. Guided Attention U-Net and a channel attention block designed by GdAlgo team.

Team GdAlgo proposed a Guided Attention U-Net (GAU-net) model demonstrated in Fig. 9. The standard convolutional layers in this model are replaced by channel attention blocks to get better visual and fidelity results. Inspired by the Multi-Stage Image Restoration approach [63], the authors adopted the corresponding multi-stage learning strategy. The model was trained to minimize *Charbonnier* loss function, the edge loss was discarded as no improvement was observed in the experiments.

5. Additional Literature

An overview of the past challenges on mobile-related tasks together with the proposed solutions can be found in the following papers:

- Image Denoising: [1, 3]
- Learned End-to-End ISP: [31, 35]
- Perceptual Image Enhancement: [34, 28]
- Bokeh Effect Rendering: [27, 33]
- Image Super-Resolution: [34, 46, 9, 58]

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